

I. STRATEGY OVERVIEW

Project BWC is a regime-aware quantitative portfolio tracking engine. Built to test structural alpha decay, hierarchical risk parity bisection, and topological model fusion on out-of-sample data. It employs strictly mathematical implementations (DuckDB, Polars, GraphicalLassoCV) to observe algorithmic behavioral bypass and track geometric degradation natively terminating in May 2026.

Report generated: 2026-05-23 06:45 UTC
Initial capital: \$1,000,000 | Benchmark: SPY

II. RISK & RETURN ANALYTICS

Metric	Portfolio	Benchmark (SPY)
Annualized Sharpe	1.30	0.95
Annualized Sortino	3.44	1.30
Calmar Ratio	6.19	1.80
Maximum Drawdown	-9.1%	-22.1%
Cornish-Fisher VaR (5%)	2.55%	-3.88%
Conditional VaR (5%)	-2.01%	-4.50%
Tail Ratio	0.95	0.85
Annualized Return	+67.52%	+10.5%
Annualized Volatility	39.69%	18.0%
Total Return	+7.13%	+22.0%

III. CONFIDENCE-FILTERED SIGNAL ENGINE

The signal engine operates as a monitoring layer that gates signals through a dual-threshold filter:

- Confidence minimum: 0.65
- Fused score minimum: 0.35
- Dynamic accuracy penalty: threshold increases when historical accuracy drops below 70%, making it harder for unreliable tickers to generate actionable signals.

The engine tracks realized vs. expected returns per ticker and uses exponentially-weighted rolling accuracy to ensure only high-quality signals reach the portfolio allocator.

IV. FAMA-FRENCH FACTOR ATTRIBUTION

Factor	Loading	Interpretation
Market (MKT-RF)	1.36	Significant market exposure
Size (SMB)	0.48	Moderate small-cap exposure
Value (HML)	-0.79	Significant value exposure
Momentum (UMD)	-0.16	Mild momentum exposure
FF3 Alpha (ann.)	0.5093	R ² = 0.43
C4 Alpha (ann.)	0.5713	R ² = 0.43

V. PORTFOLIO HOLDINGS

Ticker	Name	Sector	Qty	Avg Cost	Mkt Val	Weight
SPY	SPDR S&P 500 ETF	Broad Market	173	\$674.64	\$128,995.72	12.8%
AMZN	Amazon.com	E-commerce/C	310	\$221.03	\$82,559.20	8.2%
GLD	SPDR Gold Shares	Commodities	450	\$436.80	\$186,219.00	18.4%
GOOGL	Alphabet Inc Class	Big Tech/AI	210	\$316.65	\$80,423.70	7.9%
PLTR	Palantir Technolog	AI Software	438	\$147.10	\$59,953.44	5.9%
XLU	Utilities Select S	Utilities	1200	\$45.74	\$54,420.00	5.4%
XLF	Financial Select S	Financials	700	\$51.25	\$36,358.00	3.6%
XLI	Industrial Select	Industrials	270	\$170.75	\$46,377.90	4.6%
CAT	Caterpillar Inc	Industrials	100	\$766.00	\$87,989.00	8.7%
JPM	JPMorgan Chase	Financials	110	\$307.06	\$33,701.80	3.3%
ESGU	iShares ESG Aware	Broad Market	330	\$146.90	\$53,697.60	5.3%
XOM	Exxon Mobil	Energy	305	\$154.89	\$47,250.60	4.7%
XLE	Energy Select SPDR	Energy	60	\$57.95	\$3,569.40	0.4%
IBB	iShares Biotechnol	Healthcare	120	\$167.12	\$20,254.80	2.0%
NVDA	NVIDIA Corporation	AI Infrastru	274	\$184.04	\$59,000.42	5.8%
ORCL	Oracle Corporation	Software	161	\$155.76	\$30,924.88	3.1%

VI. WALK-FORWARD BACKTEST RESULTS

Metric	naive_equal_wei	topological_hrp	delta
total_return	0.3987	0.3406	N/A
annualized_sharpe	2.7902	3.1870	N/A
annualized_sortino	4.1367	4.7970	N/A
max_drawdown	-0.0536	-0.0502	N/A